

Task 7 - Generate Employee ID and Department ID Using Sequences and Pre-Insert Trigger

Objective

To automatically generate **Employee ID** and **Department ID** using database sequences (**EMPLOYEES_SEQ** and **DEPARTMENTS_SEQ**) through a **Block-Level Pre-Insert Trigger**, while displaying the generated IDs as read-only fields.

Description

In this task, database sequences were used to generate unique values for **Employee ID** and **Department ID**. A **Pre-Insert Trigger** was created at the block level to assign the next sequence value before a new record is inserted into the database. The ID fields were configured as **Display Items** to prevent users from entering or modifying the values manually.

Steps Performed

1. Opened Oracle Forms Builder.
2. Verified the **EMPLOYEES_SEQ** and **DEPARTMENTS_SEQ** database sequences.
3. Created a **Block-Level Pre-Insert Trigger**.
4. Used the sequences to generate the next available ID values.
5. Assigned the generated values to the **Employee_ID** and **Department_ID** fields.
6. Set the ID fields as **Display Items** (read-only).
7. Compiled and executed the form.
8. Inserted new records and verified that IDs were generated automatically.

Result

The **Employee ID** and **Department ID** were successfully generated automatically using database sequences whenever a new record was inserted. Users could view the generated IDs but could not modify them.

Concepts Used

- Oracle Forms Builder
- Block-Level Trigger
- Pre-Insert Trigger
- Database Sequences
- Display Items
- Automatic Primary Key Generation

